

LABORATORY OBSERVATION / RECORD

Course: Full Stack Web Development

Course Code: 23CM4121

Experiment No.: 2

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1. TITLE

Design Webpage Using HTML5 and CSS3 (Class vs ID Selectors Demo)

2. AIM

To design a styled webpage using HTML5 and CSS3, demonstrating the application and key structural differences between Class selectors and ID selectors.

3. OBJECTIVES

To understand the syntax and purpose of ID selectors (#) versus Class selectors (.) in CSS3.

To construct a modern dashboard interface featuring structured containers like navigation headers and feature cards.

To apply internal style rules for layout background, typography, padding, margins, borders, and box shadows.

To explore reusable UI component design patterns using classes for scalable web layouts.

4. THEORY

In HTML5 and CSS3, selectors are essential mechanisms used to target specific HTML elements for visual styling:

ID Selector (#id-name): Used to target a single, unique element on a webpage. An ID name must be unique within a document. In this experiment, the ID #main-nav is assigned strictly to the top navigation header bar.

Class Selector (.class-name): Used to target one or multiple elements across the page. Classes can be reused across several elements. In this experiment, the .feature-card and .btn-primary classes are applied repeatedly to create consistent styling across multiple dashboard cards and action buttons.

5. ALGORITHM / PROCEDURE

1. Create a new HTML file.
2. Declare the HTML5 document type and configure document metadata in the <head> tag.
3. Create an internal CSS <style> block to define rules for body, #main-nav, .feature-card, and .btn-primary.
4. Create a primary navigation header using <nav id="main-nav">.
5. Construct multiple dashboard feature cards using <div class="feature-card"> containing card headings, status text, and buttons styled with .btn-primary.
6. Save the document with an .html extension.
7. Open the file in a web browser to verify the visual rendering.

6. PROGRAM

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Class vs ID Demo</title>
  <style>
    body {
      font-family: Arial, sans-serif;
      background-color: #f4f4f9;
      padding: 20px;
    }
    /* =====
    1. ID SELECTOR (Uses "#" symbol)
    ===== */
    #main-nav {
      background-color: #2c3e50;
      padding: 15px;
      border-radius: 8px;
      color: white;
      text-align: center;
    }
    /* =====
    2. CLASS SELECTOR (Uses "." symbol)
    ===== */
    .feature-card {
      background-color: white;
      border: 1px solid #e0e0e0;
      border-radius: 6px;
      padding: 20px;
      margin: 15px 0;
      box-shadow: 0 2px 5px rgba(0, 0, 0, 0.05);
    }
    /* Another class for utility buttons */
    .btn-primary {
      background-color: #3498db;
      color: white;
      padding: 8px 16px;
      border: none;
      border-radius: 4px;
      cursor: pointer;
    }
  </style>
</head>
<body>
  <nav id="main-nav">
    <strong>Dashboard Navigation Bar</strong>
  </nav>
  <div class="feature-card">
    <h3>Cloud Storage Status</h3>
    <p>Your current usage is at 45% of total capacity.</p>
    <button class="btn-primary">Manage Files</button>
  </div>
  <div class="feature-card">
    <h3>Security Firewall</h3>
    <p>All network threat management gateways are operating normally.</p>
    <button class="btn-primary">View Logs</button>
  </div>
</body>
</html>
```

7. CODE EXPLANATION

Component / Style Rule	Purpose
#main-nav ID selector	Targets the specific navigation container. Sets dark slate grey background and white centered text.
.feature-card Class selector	Applied to both dashboard card containers. Provides white background, border, rounded corners, padding, and box shadow.
.btn-primary Class selector	Styles action buttons with blue background, white text, padding, and cursor pointer.
box-shadow	Adds a subtle drop shadow to feature cards, creating visual elevation and separation from the background.
border-radius	Rounds the corners of the navigation bar, cards, and buttons for a modern interface feel.

8. TEST CASES

Test Case 1	Verify #main-nav. Expected: Dashboard Navigation Bar appears in a dark navigation bar with white centered text.
Test Case 2	Verify .feature-card on Cloud Storage Status. Expected: white background, border, rounded corners, padding, margin, shadow and styled Manage Files button.
Test Case 3	Verify .feature-card and .btn-primary on Security Firewall. Expected: same reusable class styling.

8. TEST CASE 1

Test: Verify the ID selector #main-nav.

Expected Output: "Dashboard Navigation Bar" is displayed in a dark navigation bar with white centered text.

OUTPUT

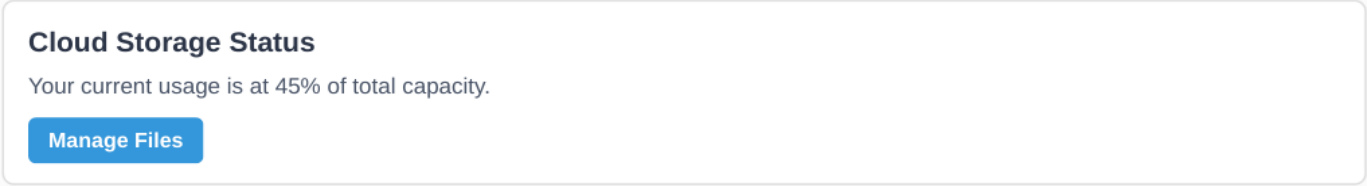


8. TEST CASE 2

Test: Verify the .feature-card class on the Cloud Storage Status card.

Expected Output: The card has a white background, border, rounded corners, padding, margin, and subtle shadow; the Manage Files button uses the primary button styling.

OUTPUT

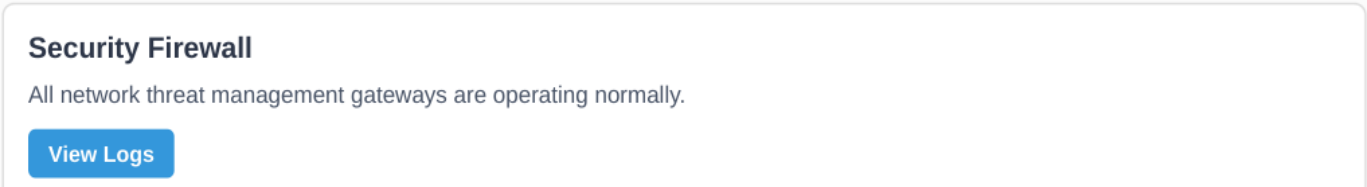


8. TEST CASE 3

Test: Verify the .feature-card and .btn-primary classes on the Security Firewall card.

Expected Output: The Security Firewall card and View Logs button are rendered using the same reusable class styling.

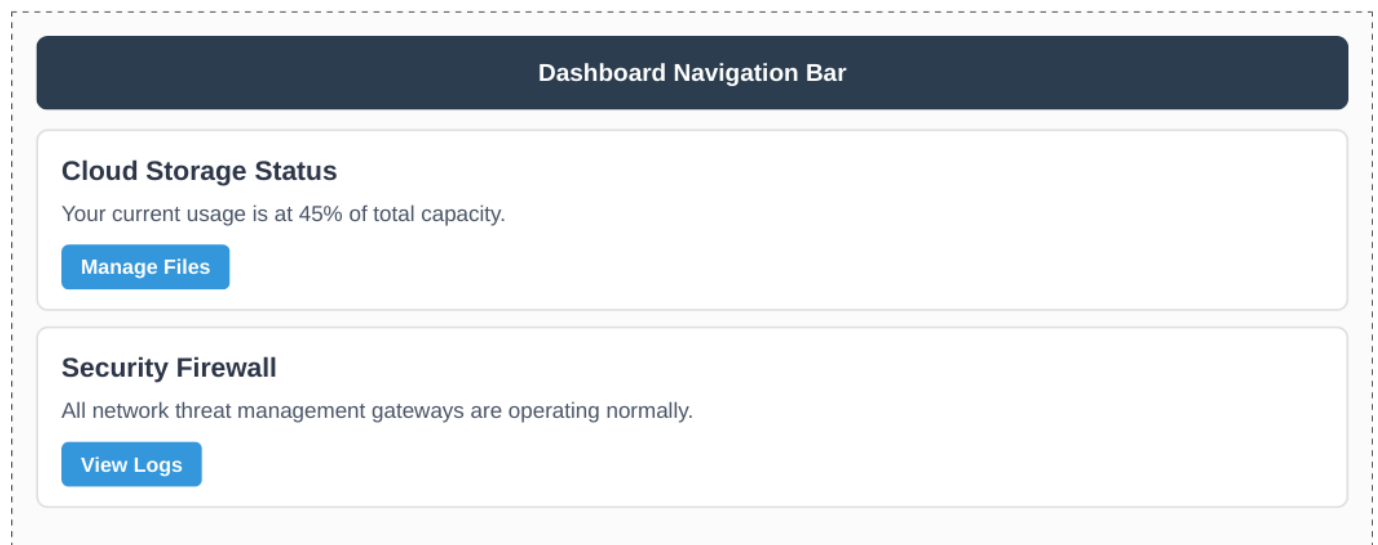
OUTPUT



9. OUTPUT

Browser Output (Screenshot / Rendering Simulation):

Browser Output (Screenshot / Rendering Simulation):



10. RESULT

Thus, a static webpage was successfully designed using HTML5 and CSS3. The webpage demonstrates the use of ID selectors for unique layout elements and Class selectors for reusable interface components, verified through web browser output rendering.